

Multi-Satellite Observation Tasks Dispatching and Scheduling

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Context and purpose : *Earth Observation Satellite (EOS)*



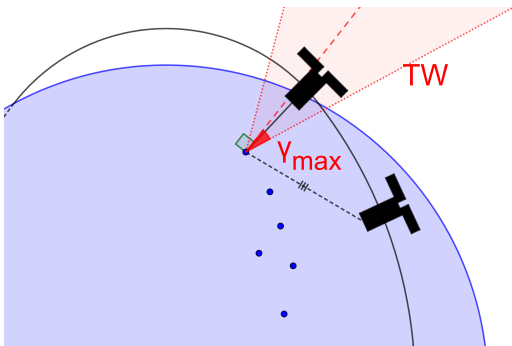
Uses : Natural hazards management, defense, mapping

Problem presentation

(i) Visibility time windows

Problem presentation

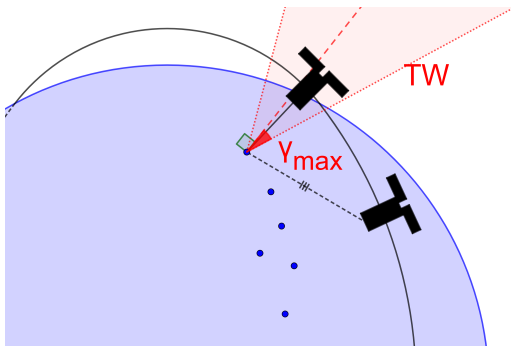
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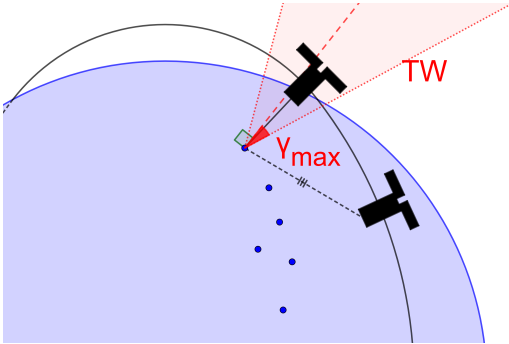
(i) Visibility time windows

(ii) *Time-dependent* transition times

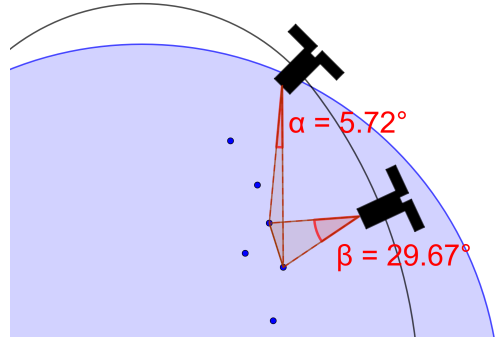


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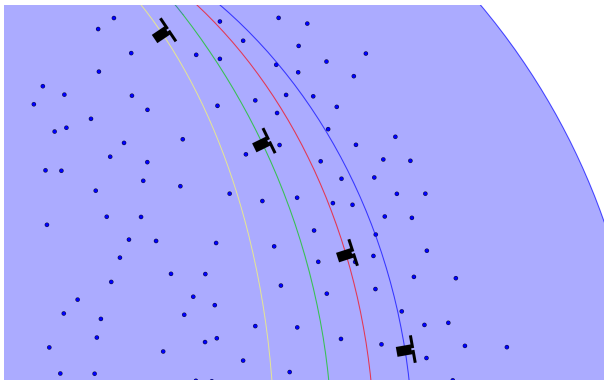
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Problem illustration



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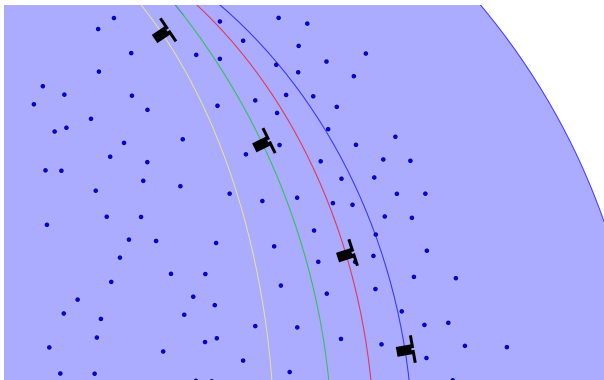


Figure: Questions : what are the right selection, dispatching and observation orders ?

Problem illustration

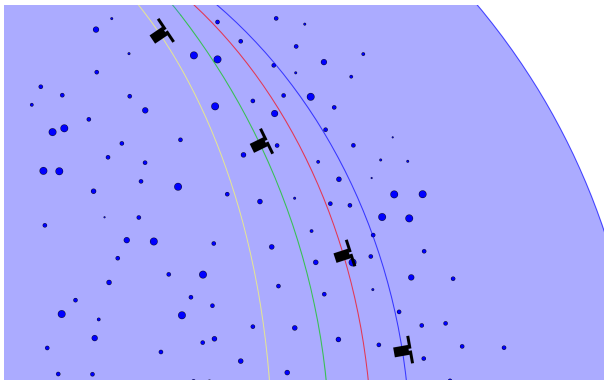


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- A time horizon H

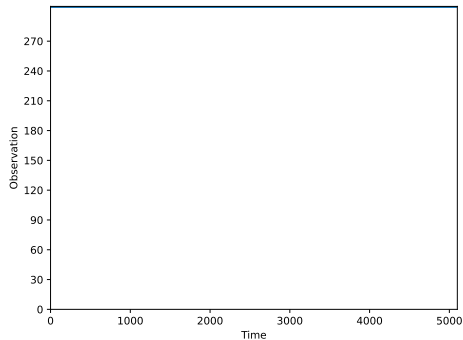


Figure: Gantt diagram

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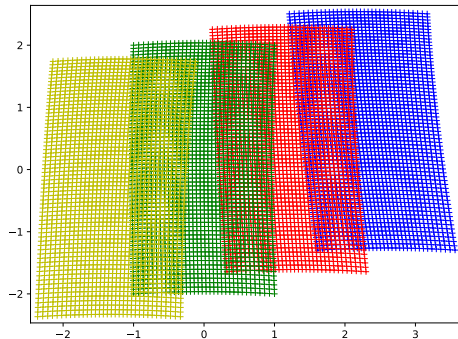


Figure: Areas overflown by the satellites

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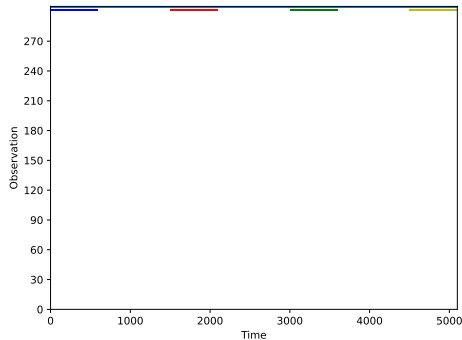


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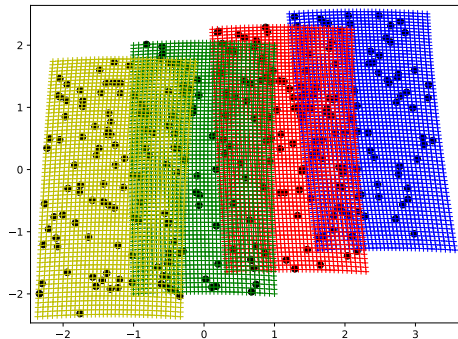


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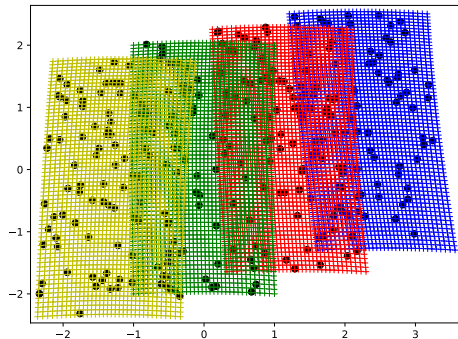


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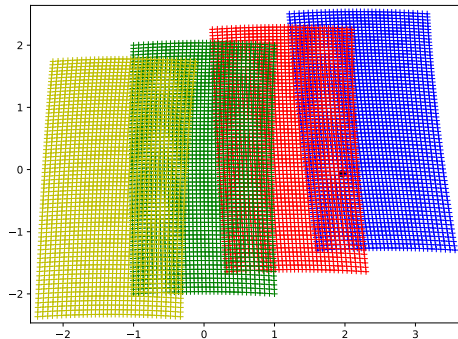


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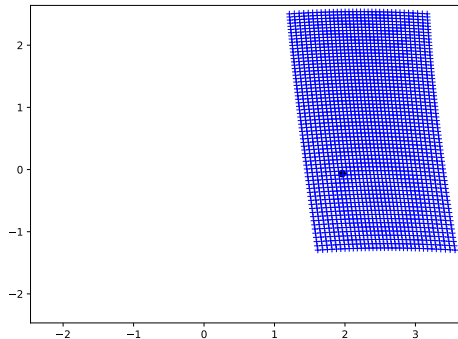


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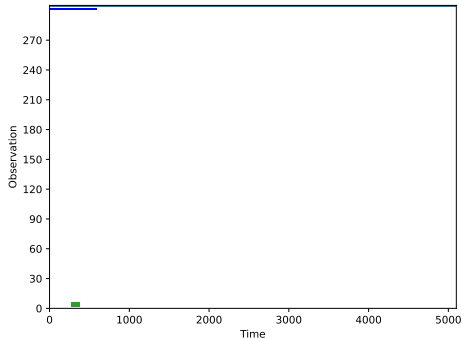


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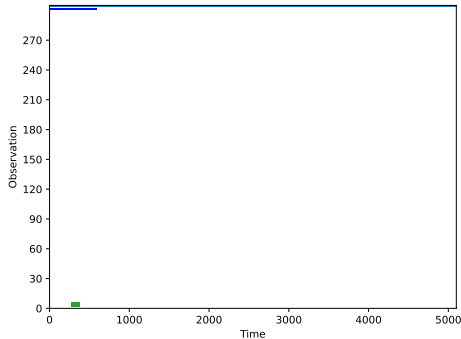


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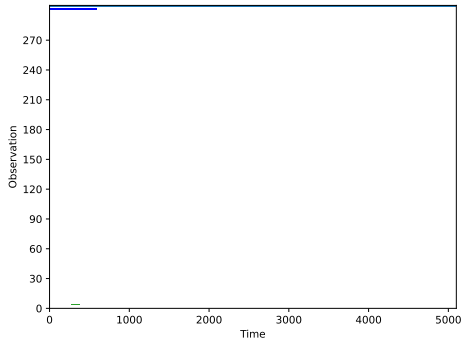


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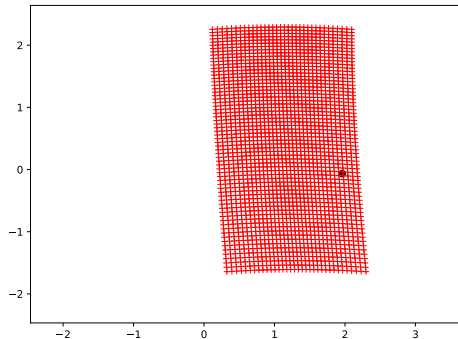


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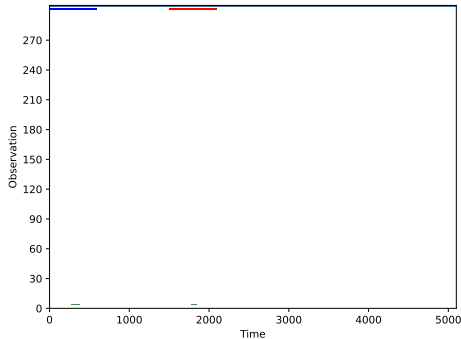


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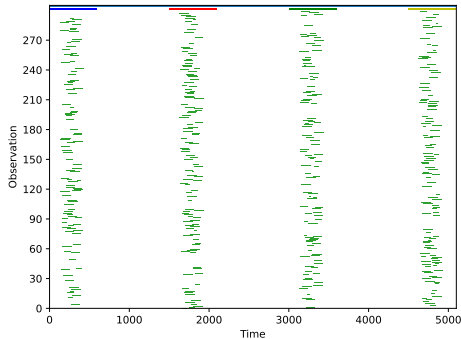


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$$\Delta_{o,o'}^s : t \rightarrow \Delta_{o,o'}^s(t)$$

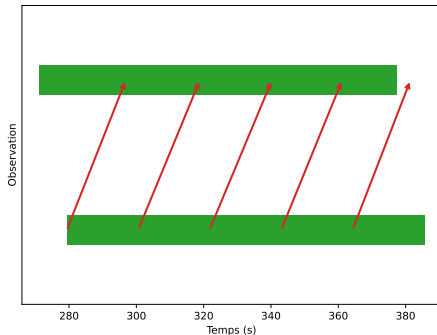


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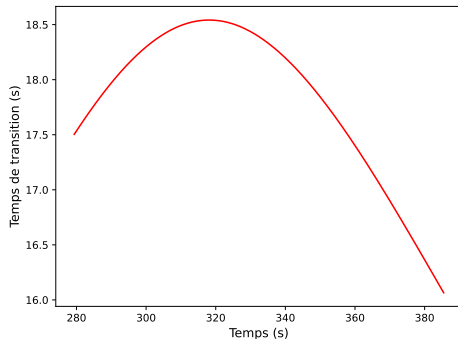


Figure: Transition time function

Modèle mathématique du problème

Définition

The *Multi-Satellite Earth Observation Scheduling Problem* (MSEOSP), for $\zeta = |\mathcal{S}|$ satellites, consists in finding ζ observation sequences $\sigma_s = [\sigma_{s,1}, \dots, \sigma_{s,K_s}]$ such that

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$$t_{\sigma_{s,i+1}} \geq t_{\sigma_{s,i}} + \text{Dur}_{\sigma_{s,i}} + \Delta_{\sigma_{s,i}, \sigma_{s,i+1}}^s (t_{\sigma_{s,i}} + \text{Dur}_{\sigma_{s,i}})$$

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- the sum of collected individual rewards $\sum_{s \in \mathcal{S}, o \in \sigma_s} \rho_o$ is maximized.

Resolution algorithm

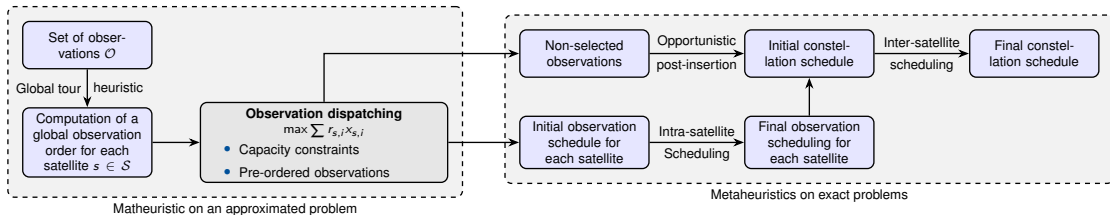


Figure: Workflow of the *MSEOSP* resolution algorithm : *Matheuristic and Metaheuristic Combination for Earth Observation tasks planning* (MAMeCEO)

Computation of a global observation order for each satellite

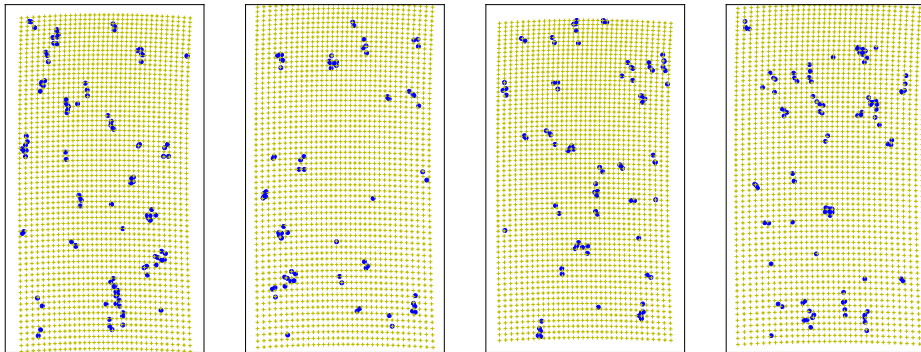


Figure: Points of view of the satellites

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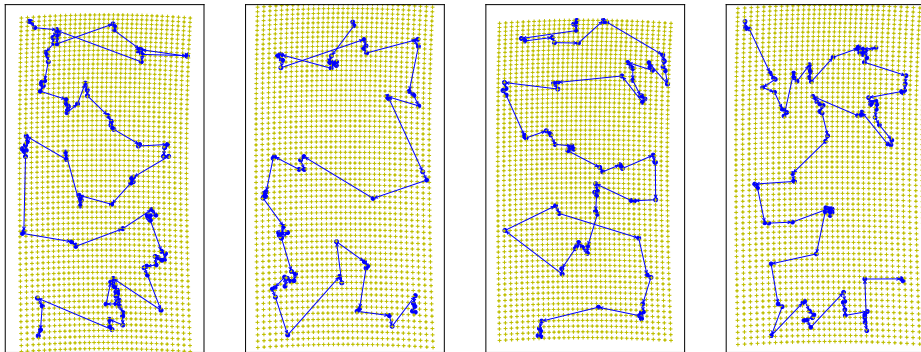


Figure: Minimal duration global observation order [1]

Selection and dispatching matheuristic

- Pre-computed global order for each satellite

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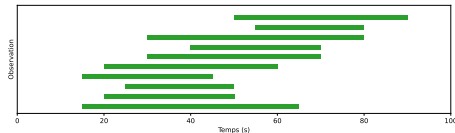


Figure: Bottom-up: pre-computed global observation order

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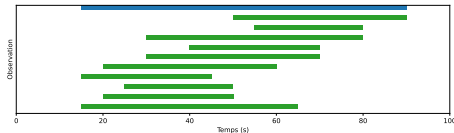


Figure: Temporal block

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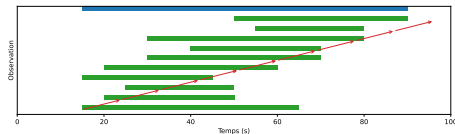


Figure: Unfeasible sequence

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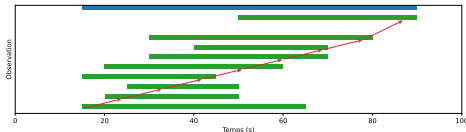


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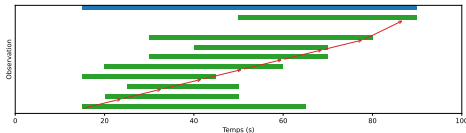


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- Individual rewards are taken into account

Matheuristic on an approximated problem

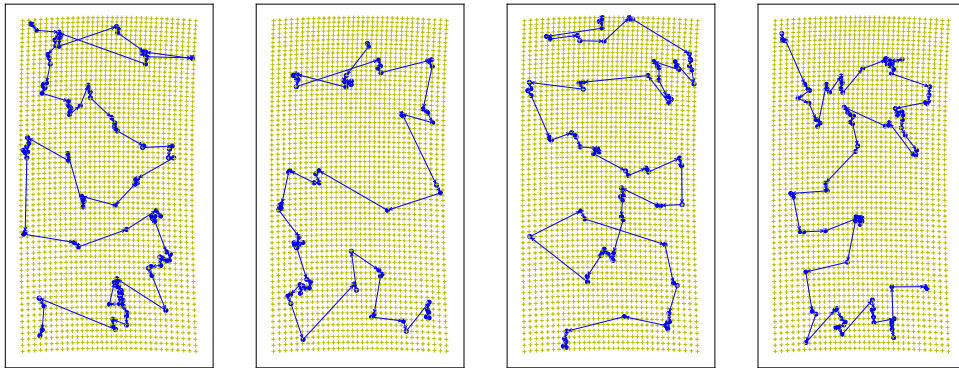


Figure: Prior to MILP resolution

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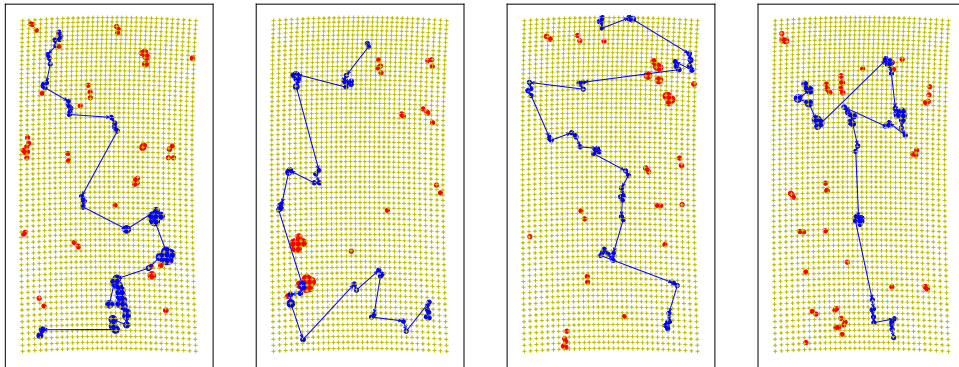


Figure: After MILP resolution

Metaheuristic on exact problems

Compliance with exact transition times and time windows

- A greedy heuristic repairs the MILP's returned solution

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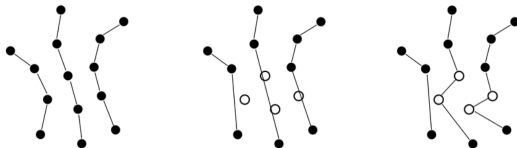


Figure: *Large Neighbourhood Search* [2, 3]

Metaheuristic on exact problems

Compliance with exact transition times and time windows

- A greedy heuristic repairs the MILP's returned solution
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- Inter-satellite planning : changing observation order and dispatching

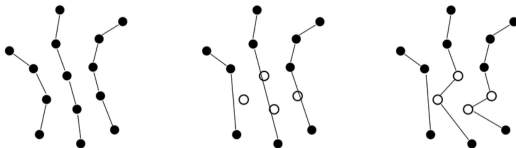


Figure: *Large Neighbourhood Search* [2, 3]

End of the algorithm

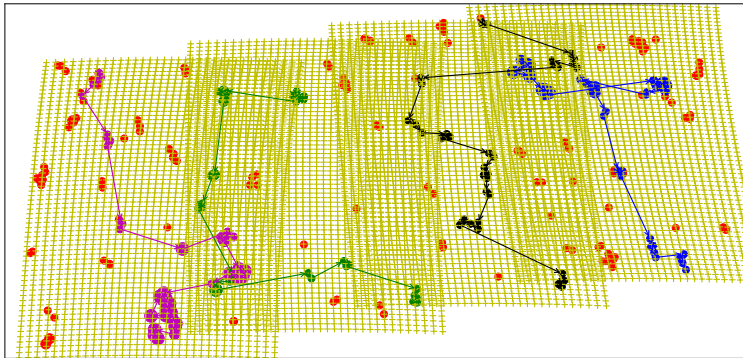


Figure: Final observation plan

Results

72 test instances : 300 POIs, 4 satellites on the same orbit approximately 25 minutes away from each other, real meteorological data from 2024 to compute individual rewards

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Algorithm	Initial solution		After intra-optimization		After inter-optimization	
	median	mean	median	mean	median	mean
Baseline LNS (20min)	5.29	6.11	n/a	n/a	0.792	1.57
Baseline LNS (1min)	5.29	6.11	n/a	n/a	1.35	1.85
MAMeCEO (1min)	10.2	10.7	4.79	5.07	0.754	1.17

Table: Optimality gaps (%) with respect to best known solutions

Conclusion and perspectives

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- Uncertainty on meteorological data

References

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**Thank you for your attention !
Any question ?**

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